King County House Sales (May–July 2014)

About the Project

This report analyzes home sales in King County, Washington, the region around Seattle.
The goal is to understand what drives home prices: how location, size and property features shape the value of a home, and how the market behaves over time.
The dashboard is designed for Cascadia Estates' management and agents, as a clear and interactive view of the local housing market.

The Data

Each record in the dataset is a single home sale in King County, between May and July 2014.

For every sale, the data includes:
- Price and sale date
- Location: city and ZIP code
- Size: living area and lot area (sqft)
- Layout: bedrooms, bathrooms and floors
- Quality: condition, view and waterfront
- History: year built and year renovated

Research Questions

Main question

Which physical and location features are linked to home prices in King County?

1. What does the market look like at a glance?
   → Market Overview

2. Where is the market most active, and where is it most expensive?
   → Location & Price

3. Which home features raise the price, and does size matter the same in every region?
   → Property Drivers

4. Is there a pattern in sales activity over time?
   → Time & Activity

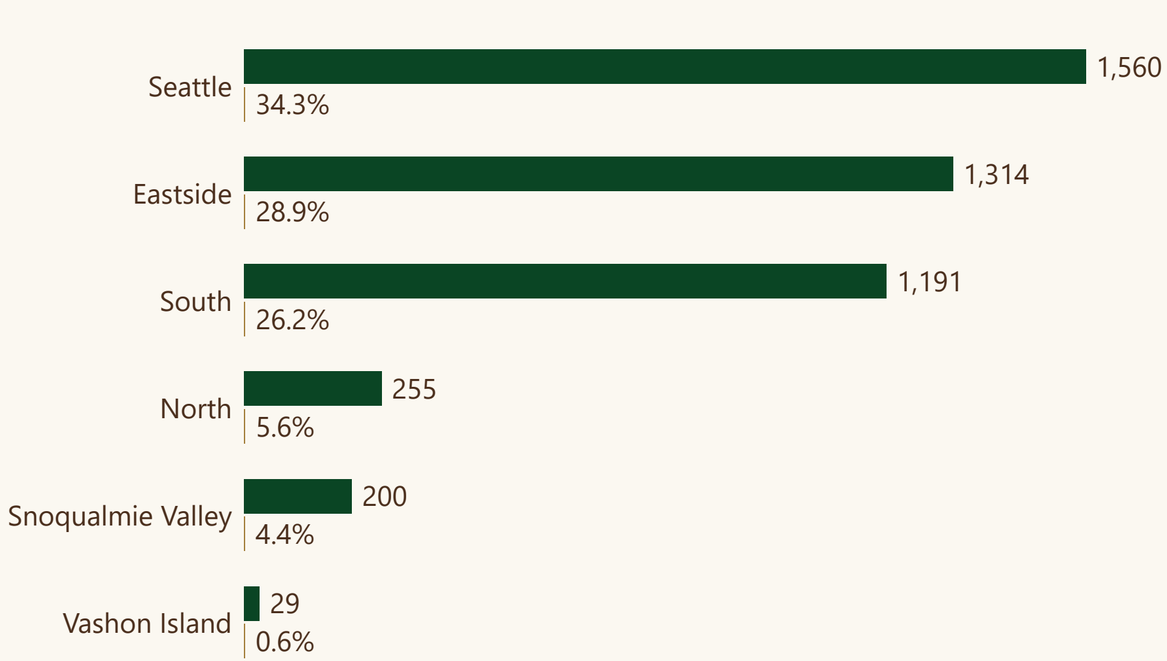
5. How much do a few extreme sales change the "typical" price?
   → Price Landscape

Market Overview

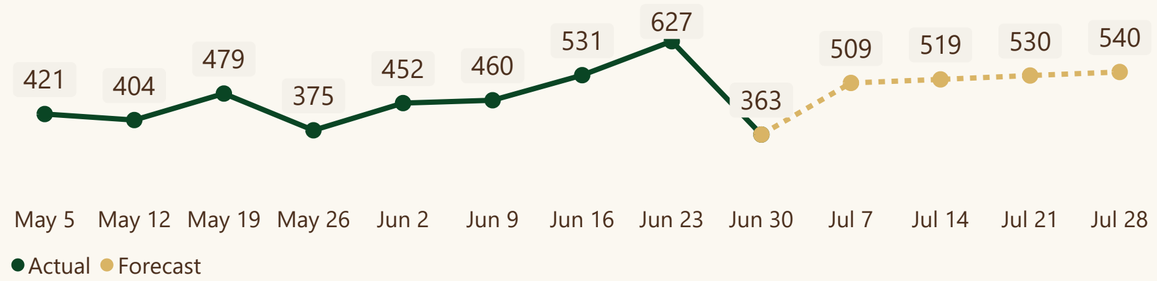
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Total Homes Sold	Sales Volume	Median Price	Median Price per SqFt	Cities
4,549	$2.54bn	$465K	$245	44

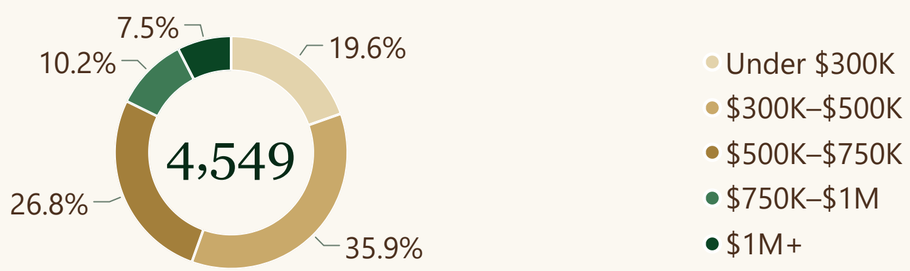
Homes Sold by Region



Weekly Homes Sold · 4-Week Forecast



Homes Sold by Price Band

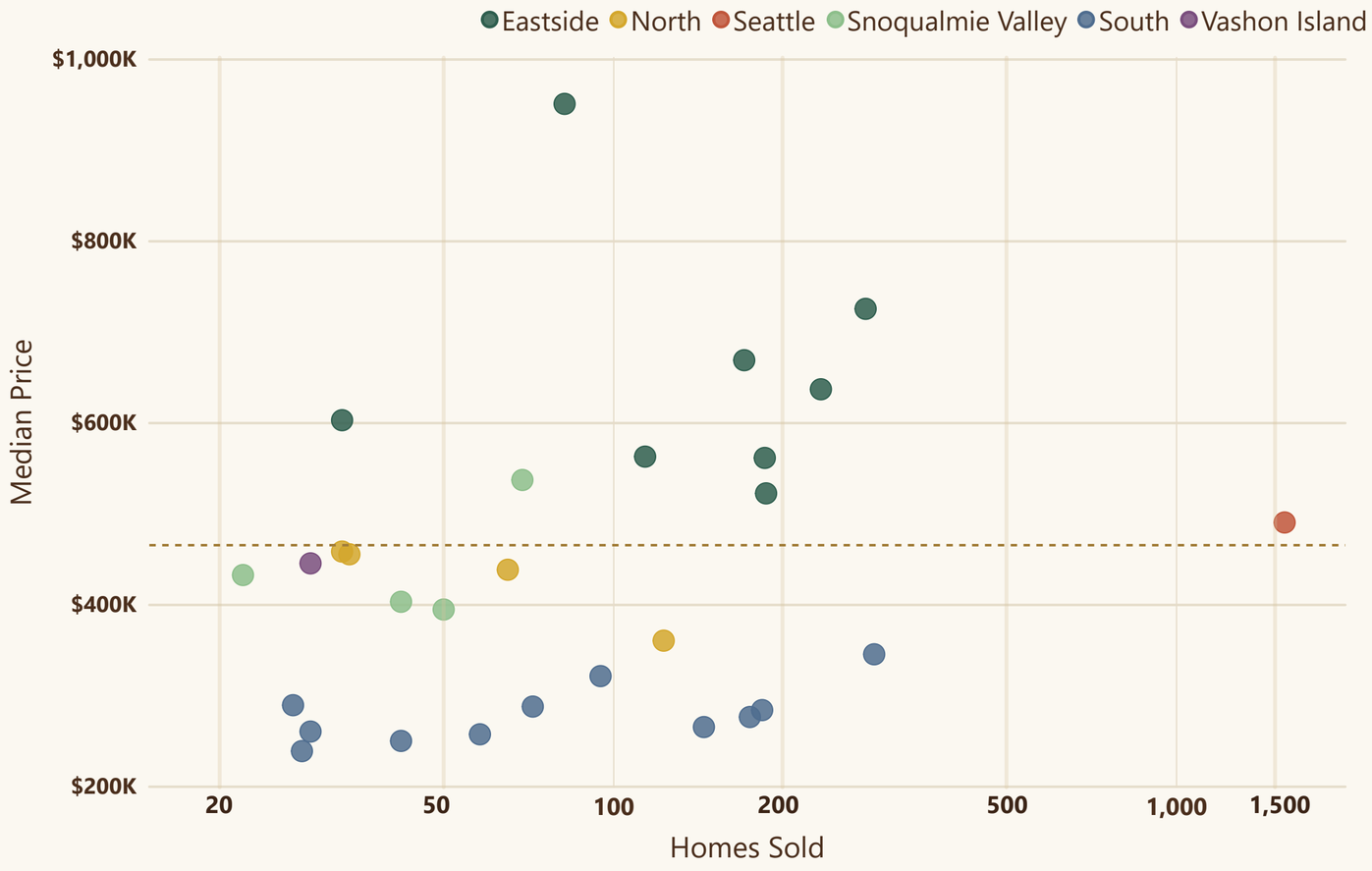


Location & Price

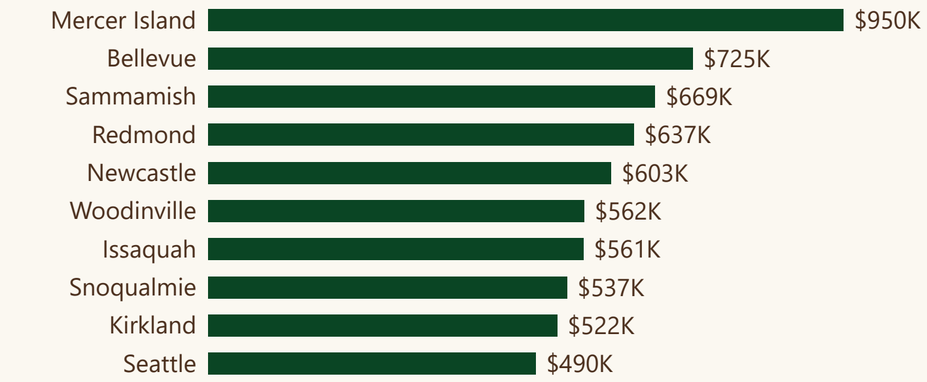
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Activity vs Price by City

Cities with 20+ sales · Dashed line: King County median ($465K)



Top 10 Cities by Median Price



Regions at a Glance

Region	Homes	Median Price	Vs County	$ / SqFt
Eastside	1,314	$643,500	38.4%	$271
Seattle	1,560	$490,000	5.4%	$313
Vashon Island	29	$445,000	-4.3%	$246
Snoqualmie Valley	200	$443,000	-4.7%	$203
North	255	$405,000	-12.9%	$226
South	1,191	$294,000	-36.8%	$161
Total	**4,549**	**$465,000**	**0.0%**	**$245**

Property Drivers

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Median price of waterfront homes

132%

higher than non-waterfront homes · county-wide

Median price of homes with a view

83%

higher than homes without a view

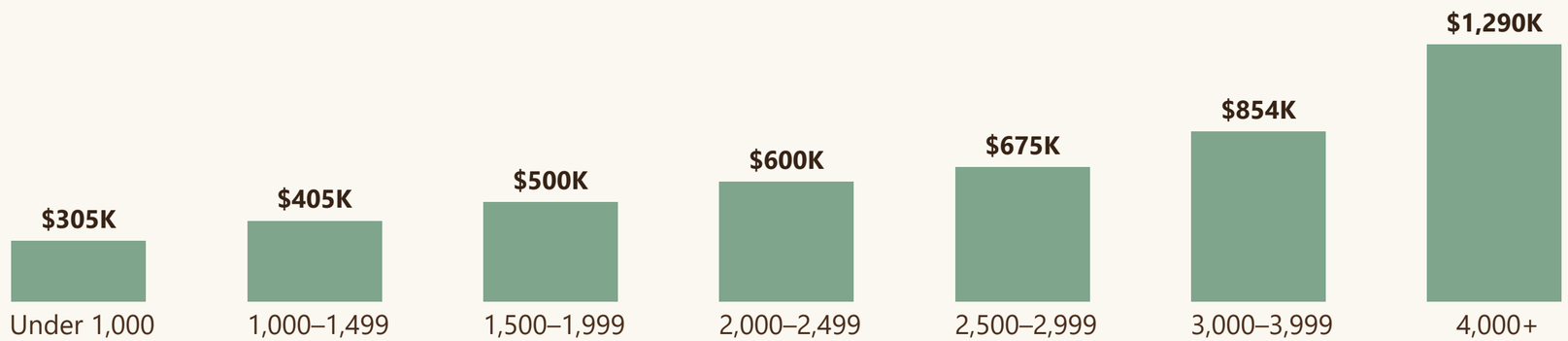
Median price of renovated homes

6%

lower than never-renovated homes

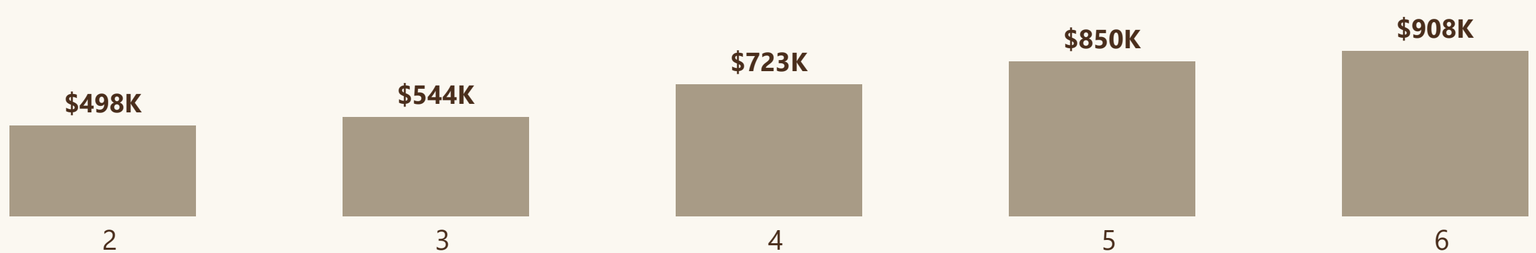
Median price by home size (sqft)

*Groups with fewer than 10 sales are hidden



Median price by number of bedrooms

*Groups with fewer than 10 sales are hidden

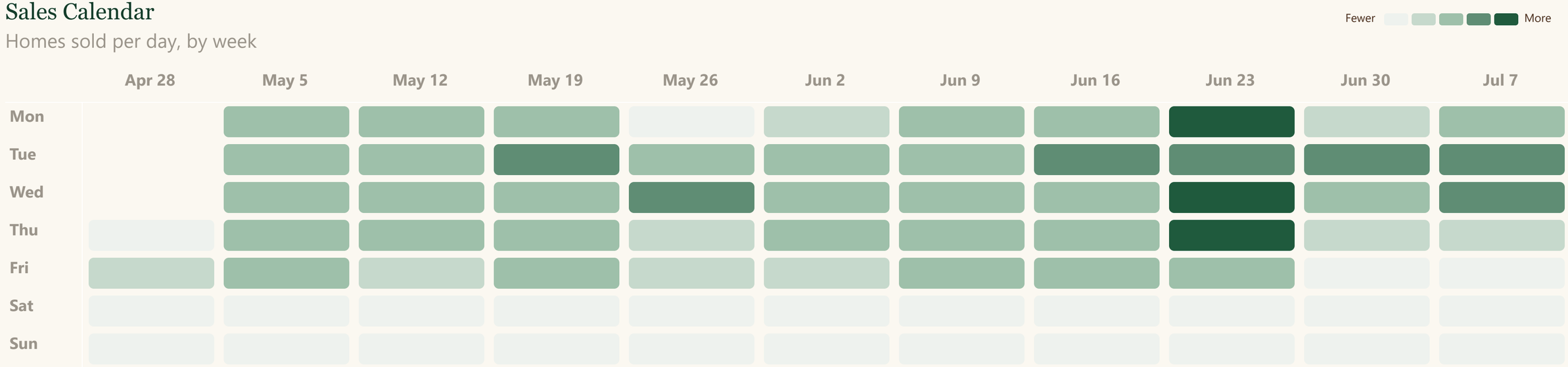


Time & Activity

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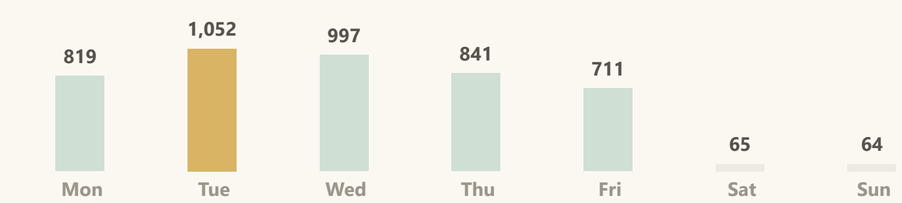
Sales Calendar

Homes sold per day, by week



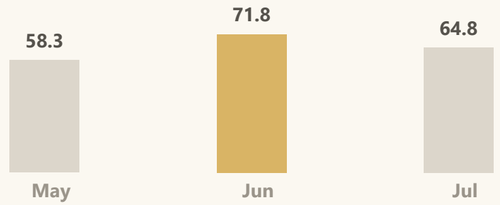
Homes Sold by Weekday

Total homes sold on each day of the week



Daily Pace by Month

Average homes sold per selling day



Selling Days

70

Weekend Sales

2.8%

Avg Sales per Selling Day

65.0

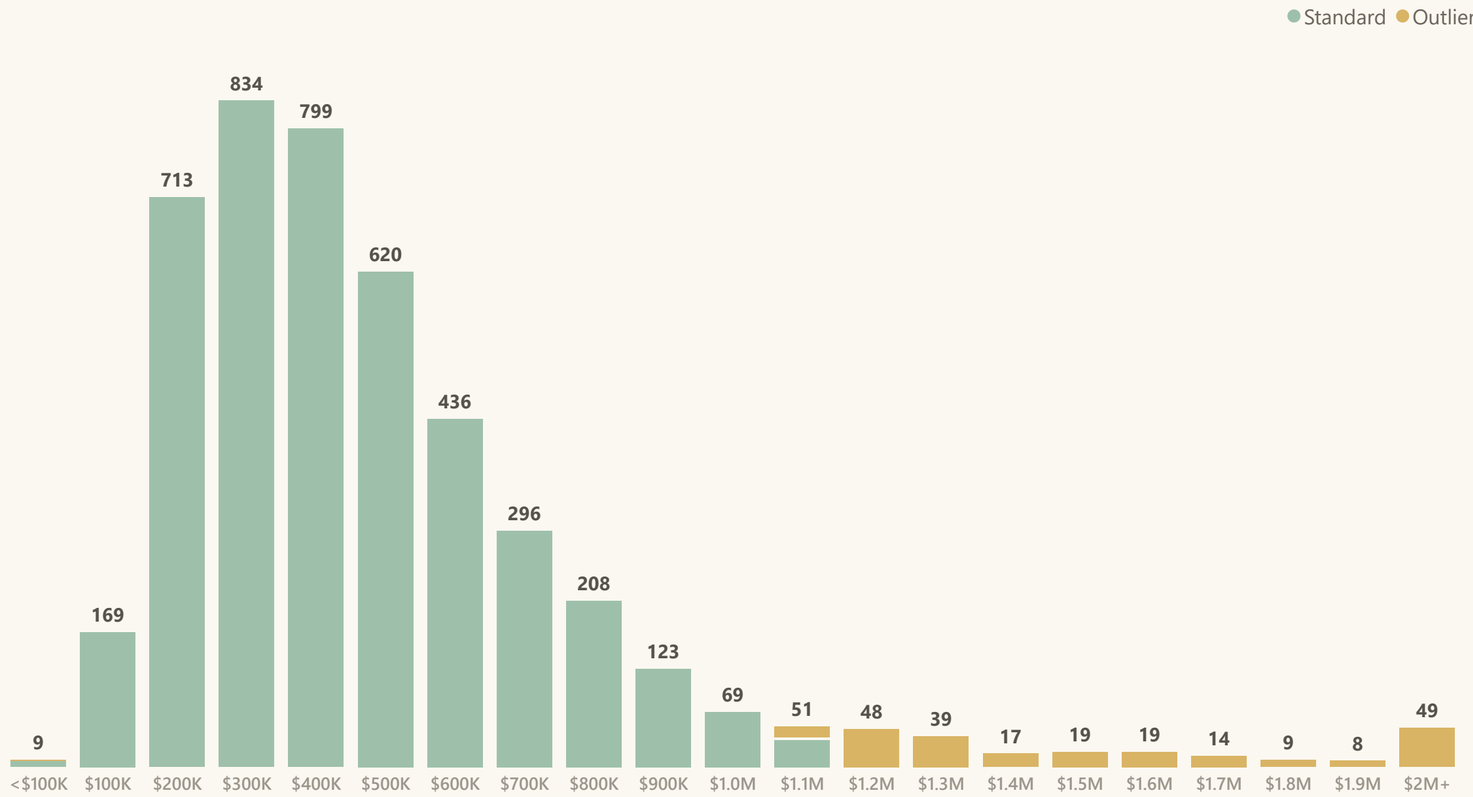
Price Landscape

Reset

All homes | Standard | **Outliers**

Price Distribution

Number of homes in each $100K price range · Outliers: above $1.15M



Standard vs Outlier Homes

Typical profile of each group

	Standard	Outlier
Median price	$450,000	$1,463,749
Median living area (sqft)	1,920	3,770
Median price per sqft	$239	$421
Avg. bedrooms	3.4	4.2
Avg. bathrooms	2.1	3.2
Waterfront share	0.4%	5.8%
Median house age	41	40

Homes in View
4,549
100.0% of all homes

Highest Sale
$26.6M
Sold in Kent

Process Notes

Data

- Source: house_sales.csv, cleaned in my Python project.
- Home sales in King County, WA (Seattle area).
- May 2 – Jul 10, 2014 · 4,549 sales.
- 44 cities, grouped into 6 regions.

Cleaning

- Removed street and ZIP (analysis is at city level).
- Removed sales with a $0 price or 0 bedrooms.
- Fixed rows where the renovation year was before the build year.
- Flagged 4 extreme deals (price over $10M or under $10K, or over 13,000 sqft) and kept them.
- Added renovation status, house age and price band.

Data Model

- Star schema: Sales (fact), Dim_Date, Dim_Location
- Many-to-one relationships, single direction
- Hierarchies: Region → City, Year → Quarter → Month → Week → Date
- Measures kept in one table, in folders by topic.



Analysis

- Medians instead of averages, because a few very expensive homes skew the average.
- Medians hidden for groups under 10 or 20 sales.
- Outliers: above $1.15M + the 4 extreme deals.
- Months compared by sales per selling day, since July is partial.
- Forecast: linear trend (LINESTX) on the 9 full weeks.

Visual Choices

- KPI cards: key numbers at a glance.
- Bar charts: rank regions and cities.
- Donut: share of each price band.
- Line: weekly trend + forecast.
- Scatter (log scale): sales vs. price by city in one view.
- Calendar heatmap: day-by-day sales pattern.
- Bookmarks as tabs and cross-filtering keep pages clean.

Key Conclusions

- Most homes sell for $300K–$500K (36%); the county median is $465K.
- Seattle is the most active region (34% of sales); Eastside is the most expensive (median $644K vs. $294K in the South).
- Size is the strongest driver: $295K under 1,000 sqft vs. $1.2M at 4,000+.
- Size works the same in every major region: homes over 2,500 sqft cost about twice as much as homes under 1,500 sqft. Location sets the base price.
- A view adds 71% to the median price; waterfront adds 132% (only 30 homes). Renovated homes sell 7% lower, since they tend to be older.
- 97% of deals close on weekdays, and June was the busiest month. The trend points to 510–540 sales per week in July.
- Outliers are only 5% of sales but 17% of sales volume. They raise the average price by 14%, while the median barely moves.